

The Ultrametric Foundation: A Mathematical Thesis on Non-Archimedean Physics

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Author: Rowan Brad Quni-Gudzinis (ORCID: 0009-0002-4317-5604)

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Abstract

This thesis develops the complete mathematical foundation for non-Archimedean physics, unifying ultrametric topology, Bruhat-Tits buildings, Berkovich analytic geometry, Tate-Amice spectral analysis, Witt vector deformation theory, and topos theory into a single framework. The central result is that the category of ultrametric spaces with contraction mappings forms an elementary topos $\mathcal{E}_{\text{ult}}$ whose internal logic encodes non-Archimedean physical law. Eight chapters traverse from Ostrowski's classification of absolute values on $\mathbb{Q}$ through Bruhat-Tits geometric realizations, Berkovich analytic spectra, p-adic Fourier analysis, Witt vector deformation theory, the ultrametric topos construction, the Hasse local-global principle as descent, and physical applications unifying adelic quantum error correction, tree-depth time, Zitterbewegung cosmology, and adelic quantum field theory.

Keywords: ultrametric spaces, Bruhat-Tits buildings, Berkovich spaces, Amice transform, Witt vectors, topos theory, Hasse principle, adelic quantum computing

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1. **Ultrametric Spaces — Complete Classification** — Ostrowski's Theorem, category $\mathbf{Ult}$, tensor product, Stone duality, p-adic numbers
2. **Bruhat-Tits Buildings as Geometric Realizations** — BN-pairs, Coxeter complexes, affine buildings, $\text{SL}(2, \mathbb{Q}_p)$ tree, classifying spaces, adelic QEC
3. **Berkovich Spaces and Analytic Geometry** — Seminorm spectra, types of points, rigid/adic/comparison geometry, sheaf theory, Berkovich-Gelfand transform, tree correspondence
4. **Tate-Amice Spectral Analysis** — Amice transform, Mahler expansion, p-adic wavelets, multiresolution analysis, non-Archimedean spectral theorem, Mahler compression as computational primitive
5. **Witt Vectors and Deformation Theory** — Witt polynomials, Teichmüller lift, deformation theory, Greenberg transform, Green functors, crystalline cohomology, prismatic

cohomology

6. **The Ultrametric Topos** — Topos construction, internal logic (Boolean with controlled AC failure), Pontryagin duality morphism, quantum logic realization, Isham-Döring program
7. **Hasse Local-Global Principle** — Classical Hasse-Minkowski, descent reformulation, Brauer-Manin obstruction, adelic cohomology, Grothendieck topology, physical examples
8. **Physical Applications — Capstone** — Kepler adelic QEC, tree-depth time, Zitterbewegung cosmology, adelic QFT, Laws of Form isomorphism

Individual chapter files available in the source repository at `ultrametric-foundation-ch{1-8}.md`.